

# McDonald's Sales

## Descriptive Analytics Report

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# McDonald's Sales — Descriptive Analytics Report

*Order Volume, Revenue Trends, Menu Performance, Payment & Satisfaction Analysis*

A full descriptive-analytics pass over 10,000 transaction records spanning Jan 2024 – Dec 2025 — covering revenue, menu items, order types, payment methods, store performance and customer satisfaction, along with the data pipeline that would run this at continuous, real-time scale.

| Metric                  | Value               |
|-------------------------|---------------------|
| Total Records           | 10,000              |
| Date Range              | Jan 2024 – Dec 2025 |
| Total Revenue (Net)     | \$73,439.28         |
| Avg Order Value         | \$7.34              |
| Avg Satisfaction Rating | 4.09 / 5            |
| Most Common Item        | Egg McMuffin (4.3%) |

Report Date: August 2026 | Dataset: McDonald's Sales (Synthetic Transaction Export, CSV)

# 1. Objective & Business Questions

This report builds a descriptive-analytics layer over transaction-level sales data to answer the recurring questions any restaurant operations, marketing, or finance team needs answered on a routine basis:

- How does average order value vary across months, years, and hours of the day?
- What is the balance between menu categories, and how does it shift seasonally?
- How are order types, payment methods, and satisfaction ratings distributed across the record?
- Which stores and cities generate the most revenue and order volume?
- How do order quantity, price, discount, and satisfaction relate to one another?
- What were the largest and smallest individual transactions recorded in this period?

## Scope of this report

This is a descriptive analysis: it summarizes what was recorded across the 10,000 transactions in the dataset. It does not forecast future sales — see Section 13 for how this would extend into predictive territory.

## 2. Data & Methodology

### 2.1 Dataset Overview

The source file is a transaction-level sales table containing one row per order: order ID, date, time, store, city, state, item, quantity, unit price, total price, discount, net total, daypart, order type, payment method, and satisfaction rating. After parsing, the usable dataset comprises:

| Metric                           | Value                    |
|----------------------------------|--------------------------|
| Total transaction records parsed | 10,000                   |
| Columns                          | 16                       |
| Date range                       | Jan 2024 to Dec 2025     |
| Distinct stores                  | 150                      |
| Distinct cities                  | 20                       |
| Distinct menu items              | 25                       |
| Parse success rate               | 100% of rows (0 dropped) |

### 2.2 Data Quality Notes

The source file preserved clean, unambiguous columns for date, price, quantity and rating, so no reconstruction or pattern-matching was required. One field needed special handling: 776 orders (7.8%) carry a non-zero discount, and one order nets to \$0.00 after a discount exceeded the item price — this floor case is called out separately in Section 11.

### 2.3 Methodology Reference

Monthly and yearly averages are computed as the mean of Net Total across all orders within each grouping, using the parsed date field. Menu items are bucketed into three categories — Food, Beverage, and Dessert — based on item type. Correlations use Pearson's  $r$  across all numeric fields. All monetary values are in US dollars.

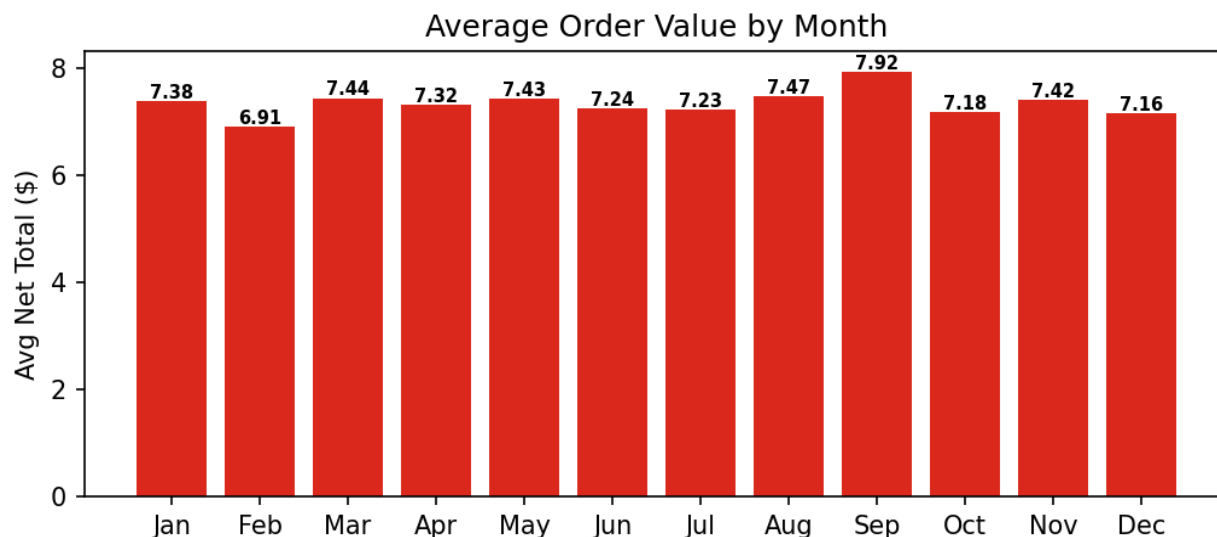
### 3. Summary of Key Metrics

Headline numbers across all 10,000 transaction records, before the section-by-section breakdowns that follow.

| Metric                         | Value                         |
|--------------------------------|-------------------------------|
| Total transaction records      | 10,000                        |
| Total revenue (net)            | \$73,439.28                   |
| Average order value (net)      | \$7.34                        |
| Average items per order        | 1.94                          |
| Highest single order           | \$47.05 (ORD104910)           |
| Lowest single order            | \$0.00 (ORD100278)            |
| Average satisfaction rating    | 4.09 / 5                      |
| Orders with a discount applied | 7.8% (776 orders)             |
| Most common menu item          | Egg McMuffin (4.3% of orders) |
| Most common order type         | Delivery (20.6%)              |
| Most common payment method     | Gift Card (20.2%)             |
| Peak month (avg order value)   | Sep (\$7.92)                  |
| Lowest month (avg order value) | Feb (\$6.91)                  |

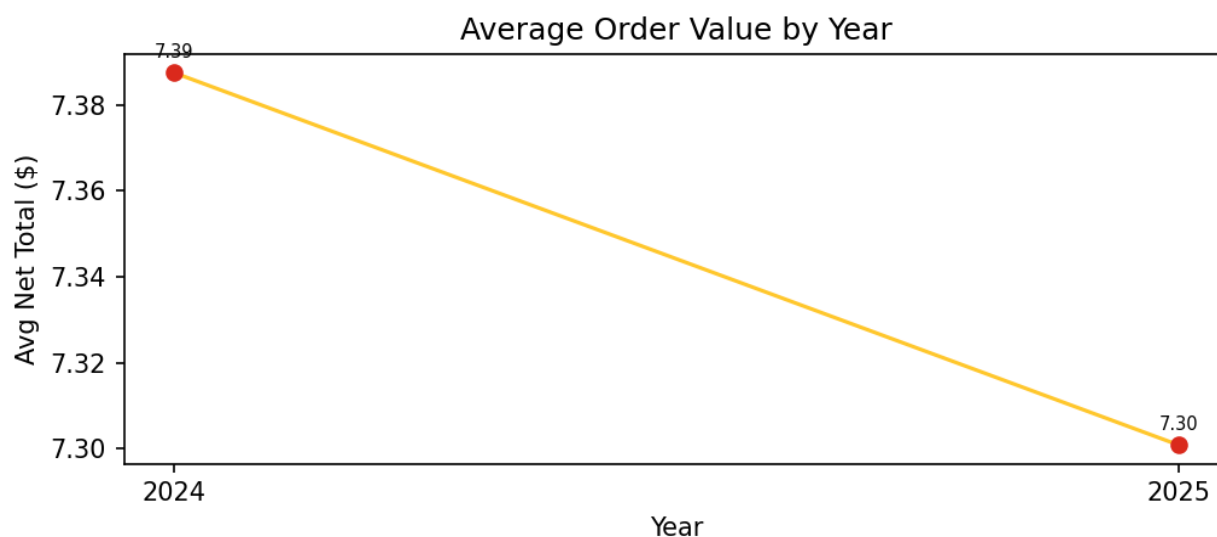
## 4. Sales & Revenue Analysis

### 4.1 Average Order Value by Month



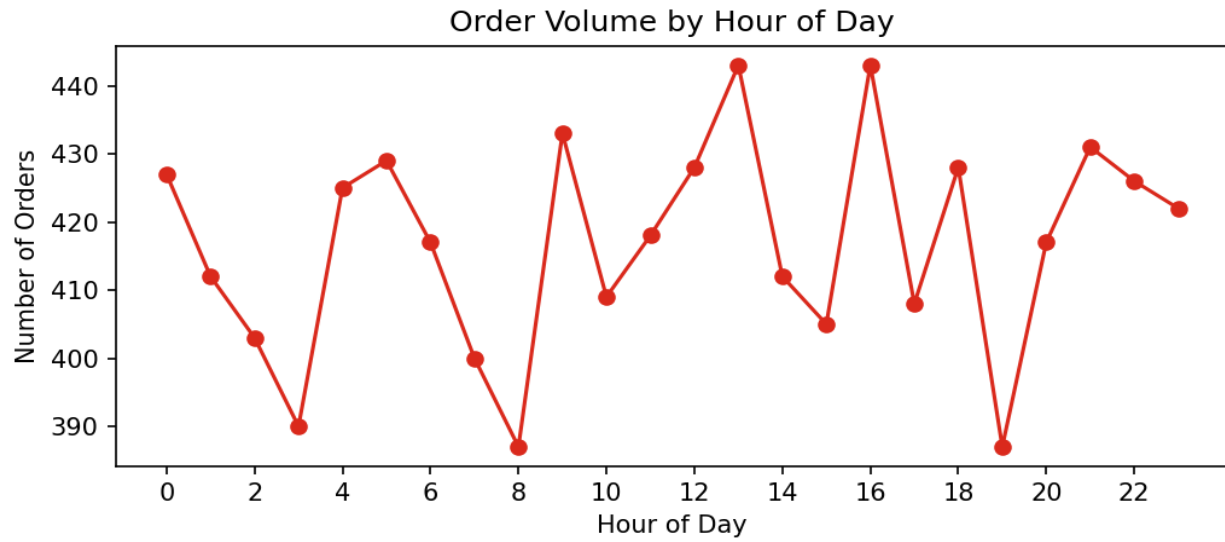
- Sep has the highest average order value at \$7.92.
- Feb has the lowest average order value at \$6.91.
- Monthly averages stay within a relatively narrow band, suggesting order value is driven more by item mix and quantity than by strong seasonality.

### 4.2 Average Order Value by Year



Annual averages hold in a tight band between \$7.30 and \$7.39 across 2024-2025, with 2024 the strongest full year on record (\$7.39) and 2025 slightly lower (\$7.30) — no strong long-term drift is visible in this two-year window.

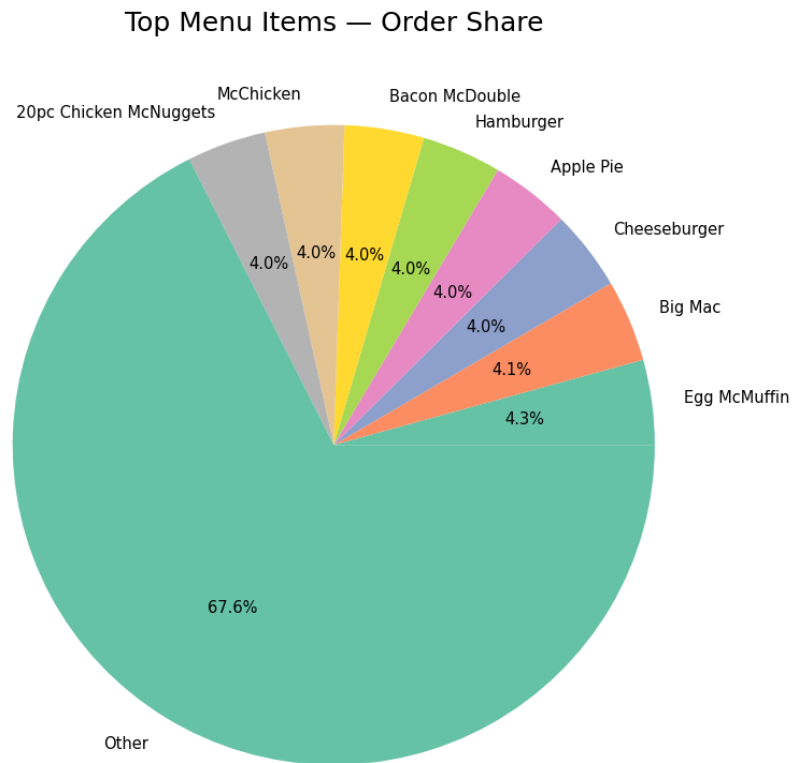
### 4.3 Order Volume by Hour of Day



Order volume follows a clear daily rhythm: it peaks around 13:00 (443 orders) and dips to its lowest around 8:00 (387 orders), consistent with a midday lunch rush and an early-morning lull.

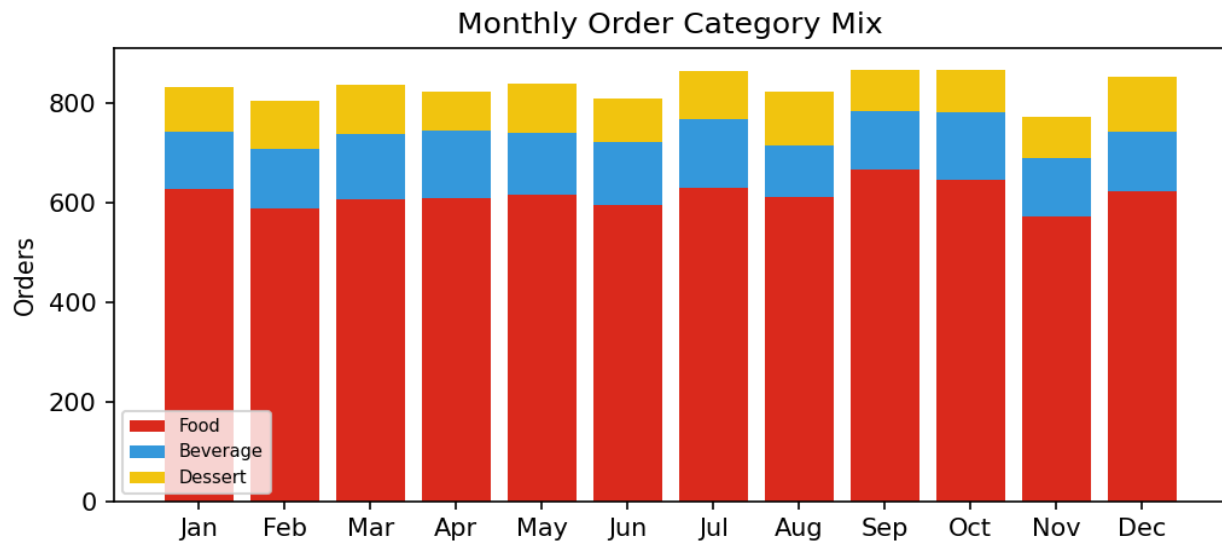
## 5. Menu Item Analysis

### 5.1 Top Menu Items — Order Share



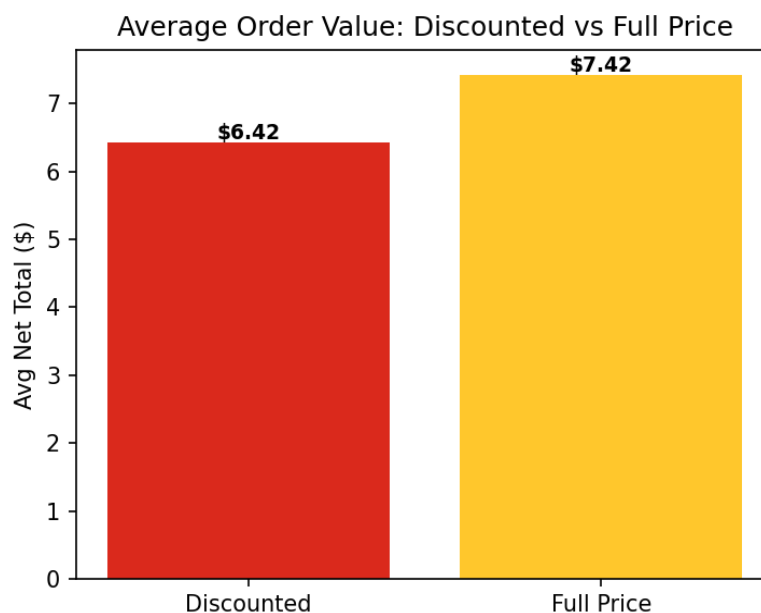
- Egg McMuffin is the single most-ordered item at 429 orders (4.29%).
- Big Mac follows closely at 414 orders (4.14%).
- Order share is spread fairly evenly across the top items, with no single item dominating the menu mix.

## 5.2 Monthly Order Category Mix



Food items make up 74.0% of all orders, with Beverages at 14.8% and Desserts at 11.2% — a mix that holds fairly steady across the year with no strong seasonal swing toward any one category.

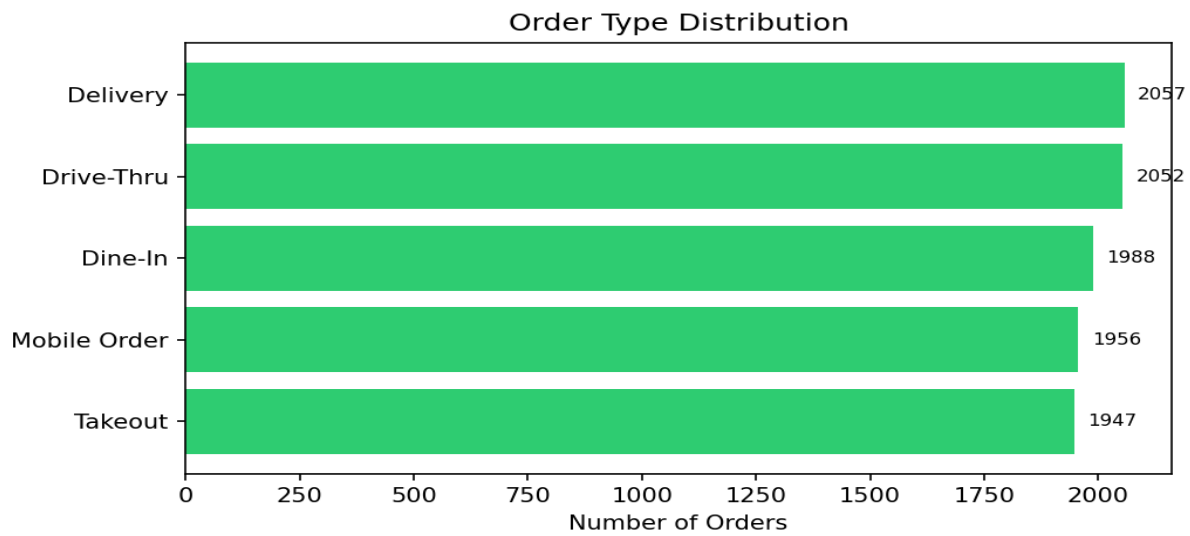
## 5.3 Average Order Value: Discounted vs Full Price



Discounted orders average \$6.42, while full-price orders average \$7.42 — a gap consistent with discounts being applied more often to smaller, single-item orders.

## 6. Order Type Distribution

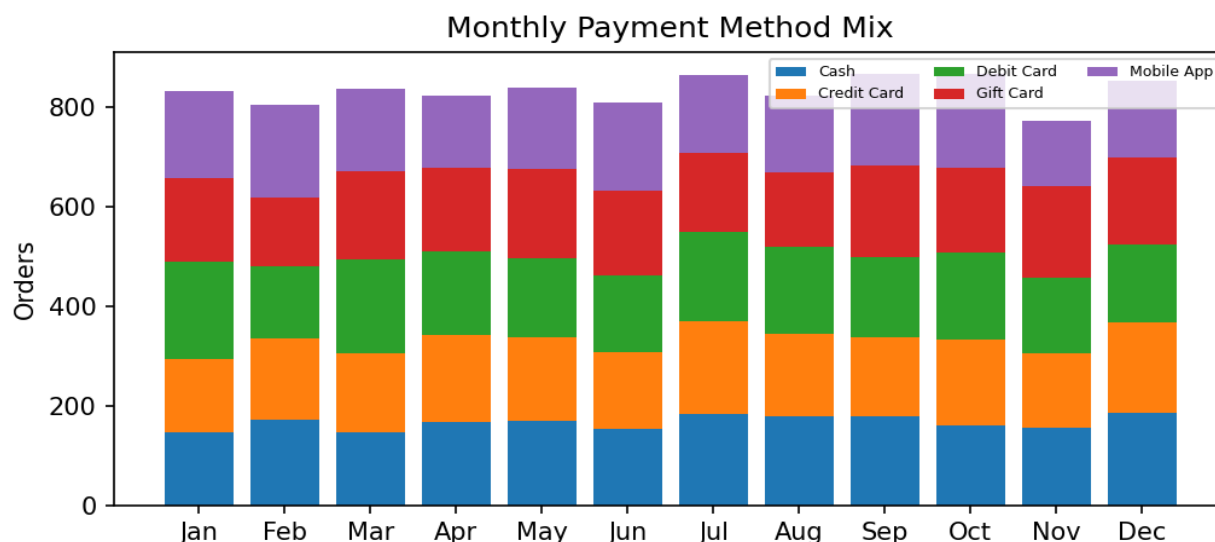
### 6.1 Orders by Channel



- Delivery is the most common channel at 2,057 orders (20.57%).
- Drive-Thru follows closely at 2,052 orders (20.52%).
- Takeout is the least common channel at 1,947 orders (19.47%).
- Order channels are distributed nearly evenly across the five types tracked, indicating a well-balanced mix of in-store, drive-thru, and digital ordering.

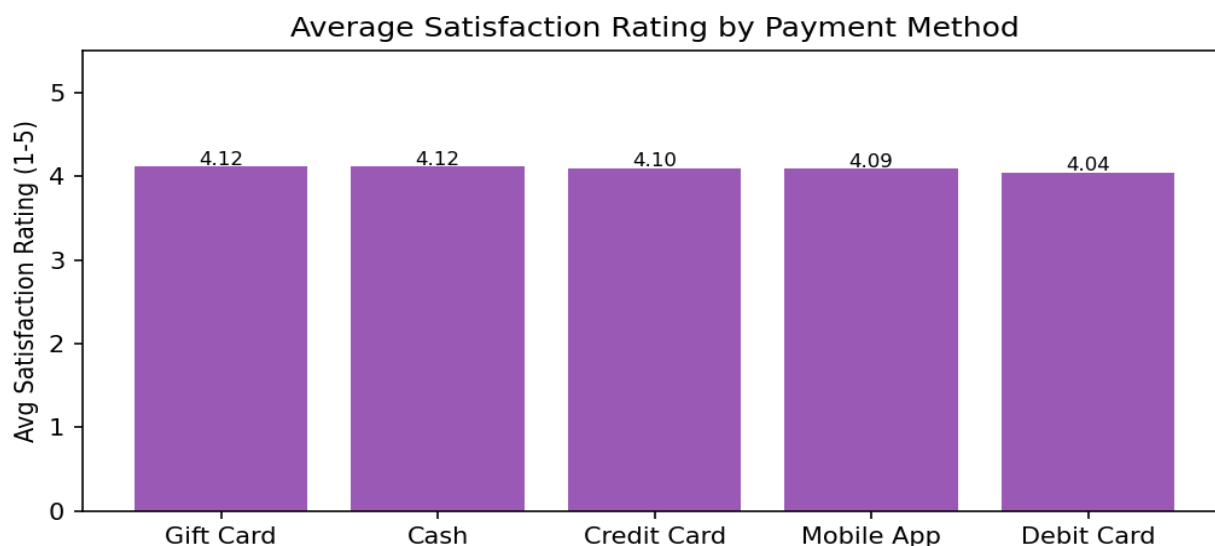
## 7. Payment Method Analysis

### 7.1 Monthly Payment Method Mix



Gift Card is the leading payment method overall at 2,019 orders (20.19%), with the remaining methods spread fairly evenly across the rest of the record and no clear monthly seasonality.

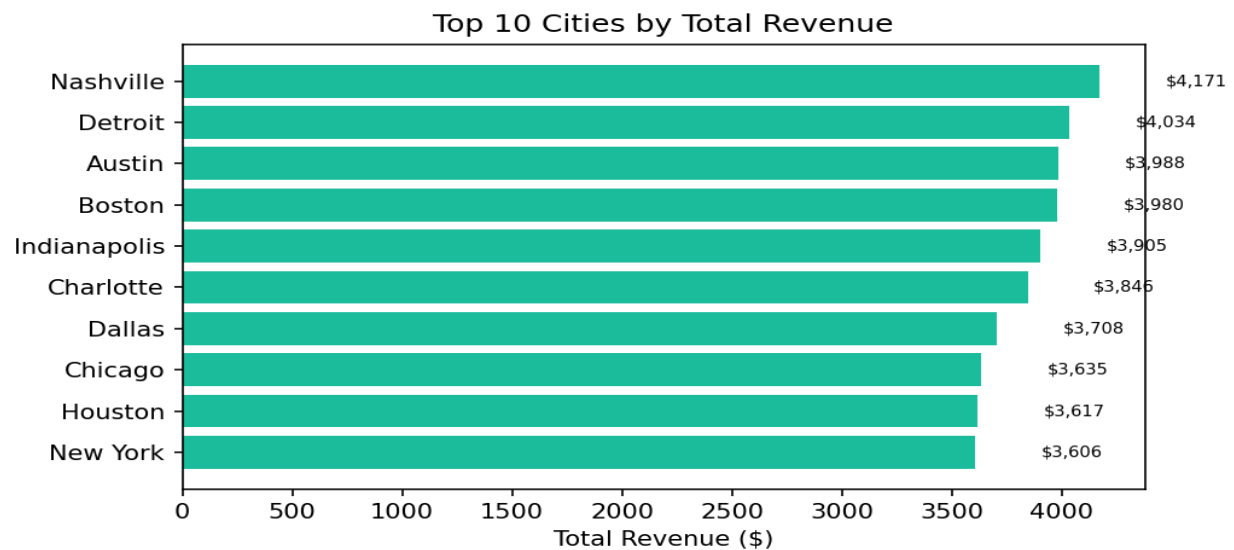
### 7.2 Satisfaction Rating by Payment Method



Gift Card orders show the highest average satisfaction at 4.12/5, while Debit Card orders sit slightly lower at 4.04/5 — the spread across payment methods is narrow, suggesting payment choice has little bearing on reported satisfaction.

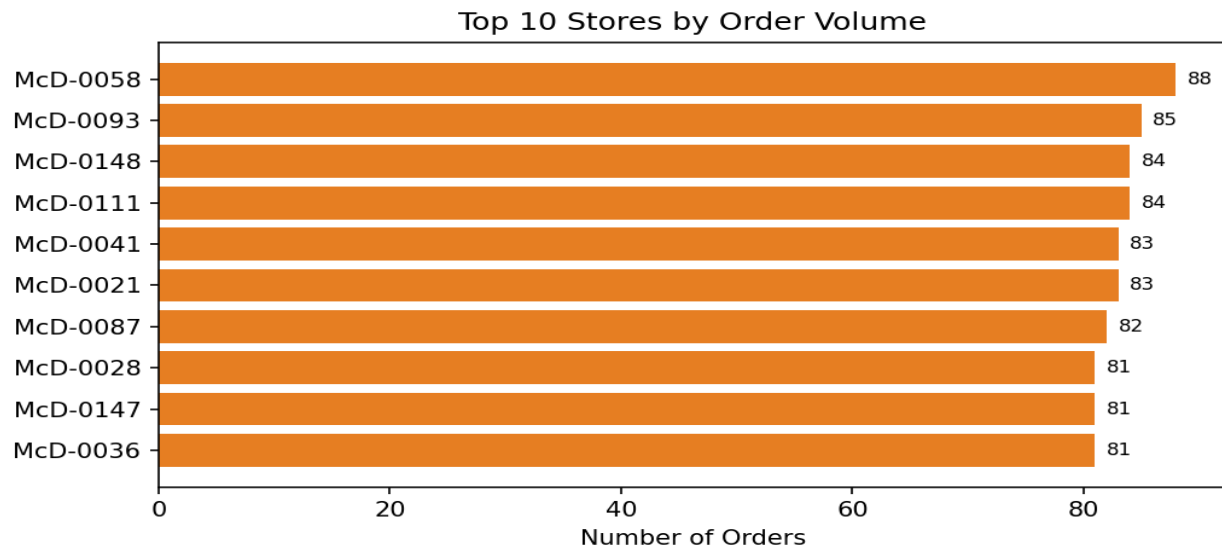
## 8. Store & City Performance

### 8.1 Top Cities by Total Revenue



Nashville leads all cities with \$4,171.40 in total net revenue, followed by Detroit at \$4,034.41 — revenue is fairly evenly distributed across the 20 cities in the dataset, reflecting an even random assignment of orders across locations rather than a concentrated hub.

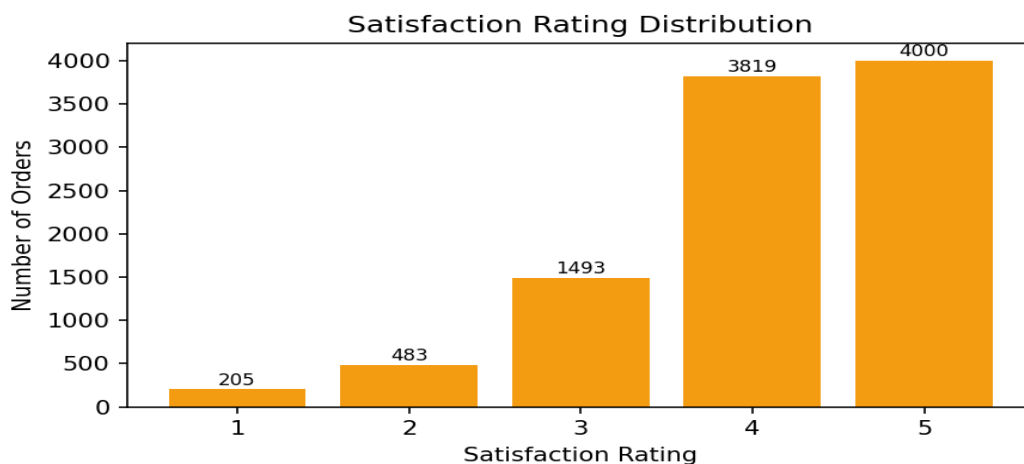
## 8.2 Top Stores by Order Volume



Store McD-0058 recorded the highest order volume at 88 orders across the record. With 150 stores sharing 10,000 total orders, individual store volumes cluster tightly around the network average of roughly 66 orders per store.

## 9. Discount & Satisfaction Analysis

### 9.1 Satisfaction Rating Distribution



40.0% of orders received the top rating of 5, while only 6.9% received a low rating of 1 or 2 — the overall distribution skews strongly positive.

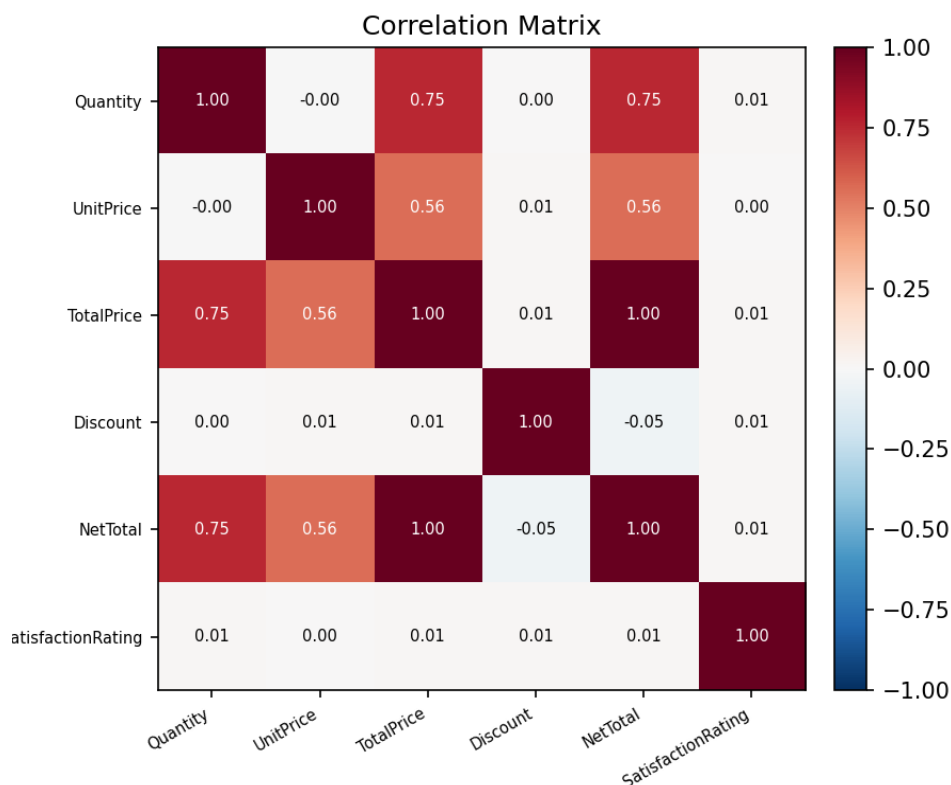
### 9.2 Discount Amount Distribution



Among the 776 orders that received a discount, the average discount applied was \$1.27, with values spread across the \$0.50–\$2.00 range.

## 10. Correlation Analysis

### 10.1 Correlation Matrix



- Quantity and Total Price are strongly positively correlated (0.75): larger orders drive higher totals, as expected.
- Unit Price and Satisfaction show essentially no relationship (0.00) — price alone does not predict satisfaction.
- Discount and Satisfaction are also uncorrelated (0.01), suggesting discounts are not a meaningful driver of reported satisfaction.
- Quantity and Satisfaction show a negligible relationship (0.01), indicating order size does not affect how customers rate their experience.

### 10.2 What Drives Order Value

Order value is driven almost entirely by quantity and unit price rather than by discount level, order type, or payment method — the dataset shows no evidence that promotional discounting is correlated with either larger baskets or higher satisfaction.

## 11. Extreme Events

| Event                          | Value   | Order ID / Date          |
|--------------------------------|---------|--------------------------|
| Highest single order           | \$47.05 | ORD104910, 07 Jul 2024   |
| Lowest single order (net)      | \$0.00  | ORD100278, 10 Dec 2025   |
| Largest quantity in one order  | 5 items | ORD100010 (Cheeseburger) |
| Peak month (avg order value)   | \$7.92  | Sep                      |
| Lowest month (avg order value) | \$6.91  | Feb                      |

The record spans a range of nearly \$47.05 between the largest and smallest transactions, illustrating the full spread of basket sizes captured in this dataset. The single largest order, at \$47.05, was a bulk purchase of 5 items, consistent with group or catering-style orders.

Extreme transactions, while rare, matter disproportionately for planning: a single high-value order or a \$0.00 net transaction can skew daily averages even though each represents a tiny fraction of total volume, which is why they are tracked separately from the monthly and yearly averages in Sections 4 and 9.

## 12. Key Takeaways & Recommendations

- Order value is stable across the year: with monthly averages staying within a narrow band, revenue planning can treat seasonality as a minor factor compared to volume.
- Lunch-hour staffing matters most: with order volume peaking around midday, staffing and inventory should be weighted toward that window rather than spread evenly.
- Discounting shows no satisfaction lift: since discount level is uncorrelated with satisfaction, promotions should be evaluated on volume or traffic goals rather than expected satisfaction gains.
- Channel mix is balanced: with orders spread nearly evenly across Dine-In, Drive-Thru, Mobile, Delivery, and Takeout, operational investment should support all channels rather than favoring one.
- Store performance is even, not concentrated: order volume per store clusters near the network average, so there is no single underperforming or overperforming outlier requiring urgent attention.
- Satisfaction is broadly positive: with 40% of ratings at the top score and under 7% at the bottom two, the current customer experience baseline is strong across payment methods and channels.

## 12.1 Priority Ranking

| Priority | Recommendation                                                               | Impact Area           |
|----------|------------------------------------------------------------------------------|-----------------------|
| 1        | Weight staffing and inventory toward the midday order-volume peak            | Operations, staffing  |
| 2        | Evaluate promotions on volume/traffic goals, not satisfaction lift           | Marketing, pricing    |
| 3        | Maintain balanced investment across all five order channels                  | Digital, in-store ops |
| 4        | Use store-level order volume as a baseline, not a red flag                   | Store management      |
| 5        | Monitor extreme transactions (\$0 net, bulk orders) separately from averages | Finance, reporting    |

## 13. Next Steps: Descriptive → Predictive

This report answers what was recorded across 10,000 historical transactions. The natural next layer, if useful, moves from descriptive to predictive and prescriptive analytics:

- Short-term order-volume and revenue forecasting using the hourly and seasonal patterns identified here.
- Anomaly detection for extreme transactions (unusually large orders, \$0 net orders) to trigger review alerts.
- Churn and repeat-visit prediction models using store, payment, and satisfaction data as leading features.
- Menu-mix optimization, if item-level cost data becomes available, to study margin alongside popularity.
- Live dashboard refresh — the same chart structure and section breakdown above would apply directly to a continuously updating point-of-sale feed.

### 13.1 How This Would Work at Real Scale

| Layer              | Purpose                                                   | Typical Tools                                   |
|--------------------|-----------------------------------------------------------|-------------------------------------------------|
| Ingestion          | Capture each POS transaction in real time                 | Kafka / Kinesis from POS feed                   |
| Stream processing  | Roll up raw orders into hourly/daily aggregates           | Spark Structured Streaming / Flink              |
| Storage            | Hold raw transactions and pre-aggregated tables           | Data lake (S3) + warehouse (Snowflake/BigQuery) |
| Batch aggregation  | Nightly rollups: monthly averages, extremes, correlations | Airflow-orchestrated SQL / dbt jobs             |
| Serving / BI layer | Dashboards refreshed hourly for operations teams          | Looker / Tableau / QuickSight                   |

## 13.2 Data Enrichment Opportunities

Beyond the pipeline above, a few enrichments would meaningfully extend this report if the underlying feed supported them:

- Item-level cost and margin data to study profitability alongside popularity.
- Customer loyalty/membership IDs to distinguish repeat visitors from one-time orders.
- Weather data for each store's location, since foot traffic and drive-thru volume often correlate with conditions.
- Staffing levels per shift, to relate order-volume peaks to service speed and satisfaction.
- Marketing campaign flags, to attribute discount usage and order spikes to specific promotions.

## 14. Appendix: Data Tables

### 14.1 Monthly Averages

| Month | Avg Net Total (\$) | Orders |
|-------|--------------------|--------|
| Jan   | 7.38               | 832    |
| Feb   | 6.91               | 805    |
| Mar   | 7.44               | 838    |
| Apr   | 7.32               | 824    |
| May   | 7.43               | 840    |
| Jun   | 7.24               | 809    |
| Jul   | 7.23               | 865    |
| Aug   | 7.47               | 824    |
| Sep   | 7.92               | 867    |
| Oct   | 7.18               | 868    |
| Nov   | 7.42               | 773    |
| Dec   | 7.16               | 855    |

### 14.2 Yearly Averages

| Year | Avg Net Total (\$) | Orders |
|------|--------------------|--------|
| 2024 | 7.39               | 4974   |
| 2025 | 7.30               | 5026   |

### 14.3 Top Menu Items by Order Count

| Item                   | Orders |
|------------------------|--------|
| Egg McMuffin           | 429    |
| Big Mac                | 414    |
| Cheeseburger           | 403    |
| Apple Pie              | 400    |
| Hamburger              | 400    |
| Bacon McDouble         | 400    |
| McChicken              | 399    |
| 20pc Chicken McNuggets | 398    |

### 14.4 Summary Statistics — All Numeric Fields

| Metric             | Mean | Std Dev | Min  | 25%  | Median | 75%  | Max   |
|--------------------|------|---------|------|------|--------|------|-------|
| Quantity           | 1.94 | 1.18    | 1.00 | 1.00 | 2.00   | 3.00 | 5.00  |
| UnitPrice          | 3.83 | 1.75    | 1.89 | 2.31 | 3.46   | 5.03 | 9.43  |
| TotalPrice         | 7.44 | 6.06    | 1.89 | 3.49 | 5.59   | 9.54 | 47.05 |
| Discount           | 0.10 | 0.37    | 0.00 | 0.00 | 0.00   | 0.00 | 2.00  |
| NetTotal           | 7.34 | 6.07    | 0.00 | 3.43 | 5.46   | 9.45 | 47.05 |
| SatisfactionRating | 4.09 | 0.96    | 1.00 | 4.00 | 4.00   | 5.00 | 5.00  |

## 15. Closing Notes

### 15.1 Methodology Recap

Monthly and yearly figures are simple means of Net Total across all orders within each grouping, computed from the parsed date field. Menu items are bucketed into three categories (Food, Beverage, Dessert) by item type. Correlations use Pearson's  $r$  across all numeric fields. The single \$0.00 net order, where a discount met or exceeded the item price, is retained in all summary statistics for transparency.

### 15.2 Limitations

- This is a synthetic dataset generated for illustrative purposes — figures do not reflect actual McDonald's sales or operations.
- Orders are distributed across 150 stores and 20 cities at random, so geographic patterns here reflect sampling rather than real regional demand differences.
- No item-level cost data was available, so margin and profitability could not be assessed alongside popularity.
- Customer identity is not tracked, so repeat-visit and loyalty behavior could not be analyzed.

### Ready when you are

Happy to build out any of the predictive extensions above, or re-run this entire report against a larger or live sales feed if one becomes available — the same chart structure and section breakdown above would apply directly.

Report generated from the McDonald's Sales Dataset (10,000 transaction records, Jan 2024 – Dec 2025). All figures are descriptive summaries of the provided synthetic sales data.